SUPERSTORE SALES ANALYSIS

1. Project Overview

This project analyzes Superstore sales data to identify key performance trends across categories and regions. By leveraging SQL queries and interactive dashboards, it evaluates profitability, shipping efficiency, and year-over-year growth to provide actionable insights for strategic business optimization.

2. Dataset

Rows: 9,994 (representing individual transaction line items)

Columns: 26 (including order details, customer information, geography, and financial metrics)

3. Data Analysis by using SQL

1. Total sales by Category.

calculated total sales across three main product categories: Furniture, Office Supplies, and Technology. It identifies which category generates the most revenue, helping businesses prioritize inventory management, marketing budgets, and resource allocation to high-performing sectors.

	category	total_sales
1	Technology	836154.02
2	Furniture	741999.73
3	Office Supplies	719046.9

2. YoY growth rate of total sales for the technology category.

measured the annual percentage change in Technology sales, highlighting growth or decline over time. It helps businesses evaluate market expansion, assess the impact of new product launches, and make data-driven decisions for future investment in the technology sector.

	order_year	total_sales	growth_rate
1	2014	175278.24	NULL
2	2015	162780.77	-7.13
3	2016	226364.22	39.06
4	2017	271730.79	20.04

3. Total sales by Sub-category.

revealed revenue distribution across specific product types, pinpointing top-performers and underachievers. Businesses can use these insights to optimize product portfolios, adjust pricing strategies, and focus marketing efforts on sub-categories with the highest growth potential.

	sub_category	total_sales
1	Phones	330007.1
2	Chairs	328449.1
3	Storage	223843.59
4	Tables	206965.53
5	Binders	203412.68
6	Machines	189238.6
7	Accessories	167380.31
8	Copiers	149528.01
9	Bookcases	114879.98
10	Appliances	107532.14

(Total rows: 17)

4. Average chair quantity for orders purchasing a table.

identified the correlation between table and chair purchases by measuring the average number of chairs bought per table order. It helps businesses optimize cross-selling strategies, design bundled furniture sets and improve inventory forecasting for related products.

	avg_chair_quantity
1	3.38

5. Number of Orders that have a total sales greater than average.

identified the volume and percentage of high-value transactions that exceed the general sales average. It helps businesses understand their reliance on large orders, refine customer segmentation, and tailor loyalty programs for high-spending clients.

	num_of_order	percent_of_total
1	1379	27.53

6. Number of orders that result in a loss after applying the discount.

identified the frequency of orders that became unprofitable due to discount policies. It helps the business evaluate the effectiveness of current promotional strategies and set discount thresholds to prevent revenue erosion and ensure sustainable profitability.

	num_of_order	percent_of_total
1	2954	58.97

7. Total sales for each month.

tracked monthly revenue fluctuations, identifying seasonal trends and peak sales periods throughout the year. It enables businesses to optimize marketing schedules, manage seasonal staffing, and align inventory levels with historical demand patterns to maximize annual profitability.

	month	total_sales
1	1/2014	14236.9
2	2/2014	4519.92
3	3/2014	55691.02
4	4/2014	28295.35
5	5/2014	23648.27
6	6/2014	34595.12
7	7/2014	33946.37
8	8/2014	27909.47
9	9/2014	81777.32
10	10/2014	31453.37
11	11/2014	78628.73
12	12/2014	69545.63

(Total rows: 48)

8. Average shipping time for each ship mode.

calculated the mean duration between order and shipment dates for different delivery methods. It evaluates logistical efficiency and service reliability, allowing businesses to optimize supply chain operations, refine shipping promises to customers, and identify bottlenecks in specific fulfillment channels.

	ship_mode	average_days_to_ship
1	Same Day	0.04
2	First Class	2.18
3	Second Class	3.24
4	Standard Class	5.01

9. Ship mode that is the most popular in each region.

identified the preferred shipping method within different geographical areas based on order volume. It helps businesses optimize regional logistics, negotiate better contracts with local carriers, and tailor shipping promotions to match the dominant customer preferences in each specific region.

	region	ship_mode	total_orders
1	Central	Standard Class	715
2	Central	Second Class	224
3	Central	First Class	174
4	Central	Same Day	62
5	East	Standard Class	828
6	East	Second Class	266
7	East	First Class	233
8	East	Same Day	74
9	South	Standard Class	495
10	South	Second Class	164

10. Top 10 customers and their state by repeat purchases.

showed the most loyal customers based on their order frequency and linked them to their primary locations. It enables businesses to implement targeted retention programs, reward high value repeat buyers, and understand geographic clusters of brand loyalty to optimize regional marketing efforts.

	customer_name	state	number_of_purchase
1	Emily Phan	Illinois	17
2	Noel Staavos	Illinois	13
3	Patrick Gardner	California	13
4	Chloris Kastensmidt	Pennsylvania	13
5	Zuschuss Carroll	Oregon	13
6	Erin Ashbrook	North Carolina	13
7	Sally Hughsby	California	13
8	Joel Eaton	Tennessee	13
9	Anna Hberlin	New York	12
10	Edward Hooks	New York	12

4. Dashboard by Power BI

This interactive dashboard provides a comprehensive visualization of Superstore sales performance, highlighting key metrics like total revenue, profitability, and regional growth. It features dynamic filters for time and category, allowing users to drill down into sub-category trends and logistical efficiency. These insights empower stakeholders to monitor business health and identify data-driven opportunities for optimization.

5. Business Recommendation

1. Create a sales promotion campaign for product categories or sub-categories with low sales performance.

2. Offer attractive promotions that encourage customers to increase their spending per transaction.

3. Boost sales during the low season by introducing seasonal product collections, such as winter-themed items, to attract customer interest.